

Lecture 1: Finding Solutions

Introduction to Data Analysis Techniques

1 Data Analysis in Science

Definition 1.1 (Analytical Method and Analytical Solution). [def:analytical-method] When we use *analytical methods* to solve a problem, we are manipulating the mathematical expressions until unknowns are being isolated into a standard mathematical form.

An *analytical solution* is a solution for a problem such that the unknowns are expressed in a closed form.

Example 1.1 (Analytical Solution of a Quadratic Equation). [eg:quadratic-equation] Consider the quadratic equation

$$ax^2 + bx + c = 0, \quad (1)$$

where a , b , and c are constants. The analytic solution for x can be found using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (2)$$

Definition 1.2 (Numerical Method and Numerical Solution). [def:numerical-method] *Numerical methods* are methods to obtain numerical solutions.

A *numerical solution* is a solution for a problem such that the unknowns are expressed in an numerical form.

Example 1.2 (Numerical Solution of a Trigonometric Equation). [eg:trig-equation] Consider the trigonometric equation

$$x = \beta \cos x \quad (3)$$

where β is a constant.

It is very difficult to find an analytical solution for this equation using algebraic manipulations. However, we can use numerical methods to find an approximate solution for specific β values.

The numerical solutions can be

$$\beta = 1, \quad x \approx 0.73909, \quad \beta = 0.5, \quad x \approx 0.4508$$

Remark 1.1 (Advantages and Disadvantages of Numerical Methods). [rem:numerical-methods]

1. For numerical methods, we must have all parameters of the problem known.
2. If parameters change, solution must be re-done.
3. Solutions may be not exact.
4. Numerical solutions often considered inferior to analytic solutions, if exist. *ie, we should use analytical methods whenever possible or too difficult.*

2 Using Taylor Series to Approximate Functions

Definition 2.1 (Taylor Series). [def:taylor-series] For a function $f(x)$ that is infinitely differentiable at a point a , then:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Remark 2.1 (Taylor Series Approximation). [rem:taylor-series-approx] It is impossible to compute the infinite sum in practice. When we find the analytical solution of a problem, we can use the first few terms of the Taylor Series to approximate the function.

The more terms we use, the better the approximation.

Example 2.1 (Electric Field of a Dipole for $x \gg a$). [eg:dipole-field] Consider the electric field on the axis of a dipole with charges $+q$ at $x = a$ and $-q$ at $x = -a$:

$$E = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(x-a)^2} - \frac{1}{(x+a)^2} \right]$$

Take the Taylor series expansion of E for $x \gg a$:

$$E = \frac{q}{4\pi\epsilon_0 x^2} \left[4\frac{a}{x} + 8\left(\frac{a}{x}\right)^3 + 12\left(\frac{a}{x}\right)^5 + \dots \right]$$

Keeping the first two terms gives an approximation for $x \gg a$:

$$E \approx \frac{q}{4\pi\epsilon_0 x^2} \left[4\frac{a}{x} + 8\left(\frac{a}{x}\right)^3 \right]$$

Which means, for $x \gg a$, $E \propto x^{-3}$.

3 Fixed Point Iteration Method

Definition 3.1 (Fixed Point Iteration Method). [def:fixed-point-iteration-method] The idea for *fixed point iteration method* is to first find a function $g(x)$ such that root r is a fixed point of $g(x)$, i.e., $g(r) = r$

Convention 3.1 (Procedure of Fixed Point Iteration Method). [conv:fixed-point-iteration] The classical procedure of fixed point iteration method is as follows:

1. Find a function $g(x)$ such that root r is a fixed point of $g(x)$, i.e., $g(r) = r$.
2. Choose an initial guess x_0 .
3. Choose a maximum number of iterations N
4. Iterate $x_{n+1} = g(x_n)$ for N times

Remark 3.1 (Error Analysis of Fixed Point Iteration Method). [rem:fixed-point-iteration-error] By using this method, we usually believe that the root, r , is close to the initial guess, x_0 . Also, it should be within x_n and x_{n+1} for all n .

Which means, the error of the n -th iteration is $|x_n - r| \leq |x_{n+1} - x_n|$.

Convention 3.2 (Optimized Fixed Point Iteration). [conv:optimized-fpi] We may want to set an error tolerance, tol , and stop the iteration when $|x_{n+1} - x_n| < \text{tol}$.

1. Find a function $g(x)$ such that root r is a fixed point of $g(x)$, i.e., $g(r) = r$.
2. Choose an initial guess x_0 .
3. Choose a maximum number of iterations N
4. Set an error tolerance, tol
5. Iterate $x_{n+1} = g(x_n)$
6. Check if $|x_{n+1} - x_n| < \text{tol}$, if yes, stop the iteration and return the value.
7. If the maximum number of iterations is reached, stop the iteration and return the value.
(For some algorithms, we may want to return an error message instead.)

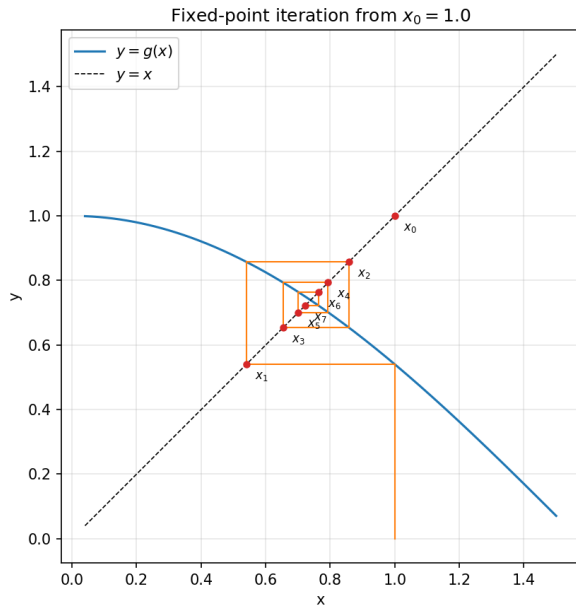
Example 3.1 (Convergent Case: Fixed Point Iteration on $x - \cos x = 0$). [eg:cos-fixed-point] Apply Optimized Fixed Point Iteration to $f(x) = x - \cos x = 0$.

1. Rewrite $f(x) = 0$ as $x = \cos x$, so $g(x) = \cos x$ and $g(r) = r$.
2. Initial guess: $x_0 = 1$.
3. Maximum iterations: $N = 100$.
4. Error tolerance: $\text{tol} \in \{5 \times 10^{-2}, 1 \times 10^{-3}\}$ (two runs for comparison).
5. Iterate $x_{n+1} = \cos x_n$.

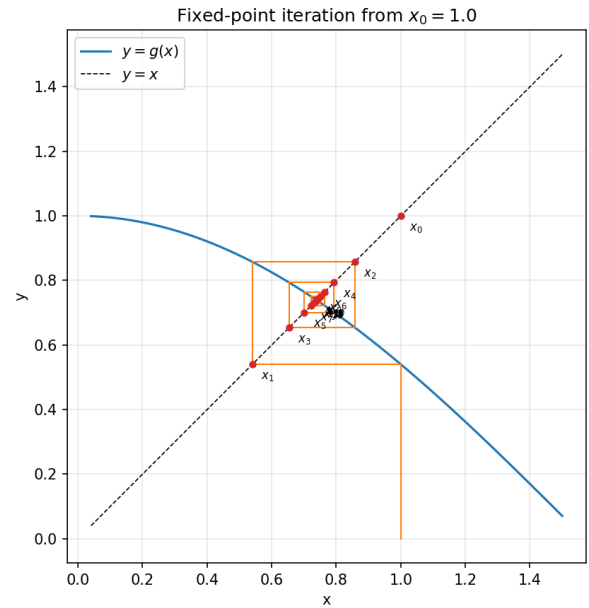
6. Stop when $|x_{n+1} - x_n| < \text{tol}$ and return x_{n+1} .

7. If n reaches N , stop and return the last x_{n+1} .

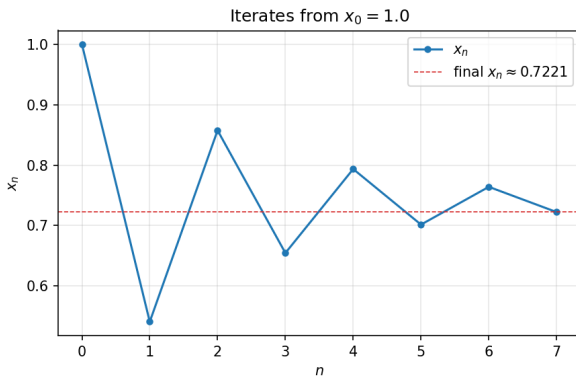
Both runs spiral into the fixed point $r \approx 0.739$; a tighter tolerance just adds more steps near r . The cobwebs (top row) show the geometric approach; the iterate-vs-index plots (bottom row) show the same convergence as damped oscillation.



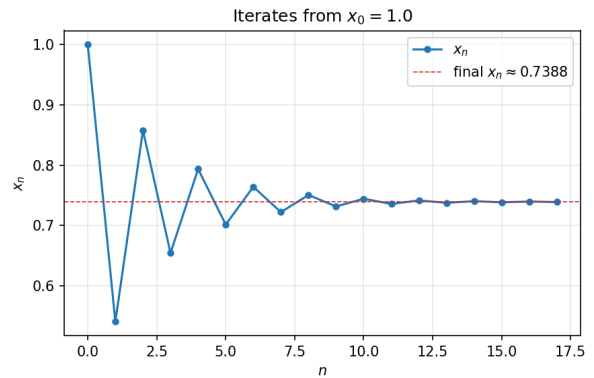
(a) Cobweb, $\text{tol} = 5 \times 10^{-2}$.



(b) Cobweb, $\text{tol} = 1 \times 10^{-3}$.



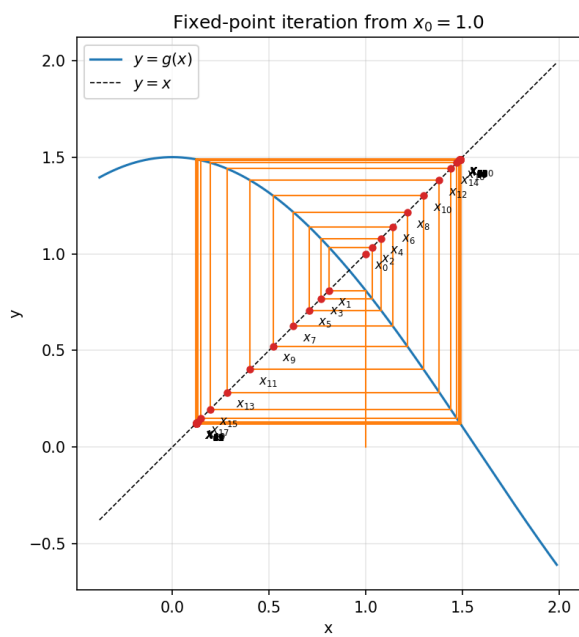
(c) x_n vs. n , $\text{tol} = 5 \times 10^{-2}$.



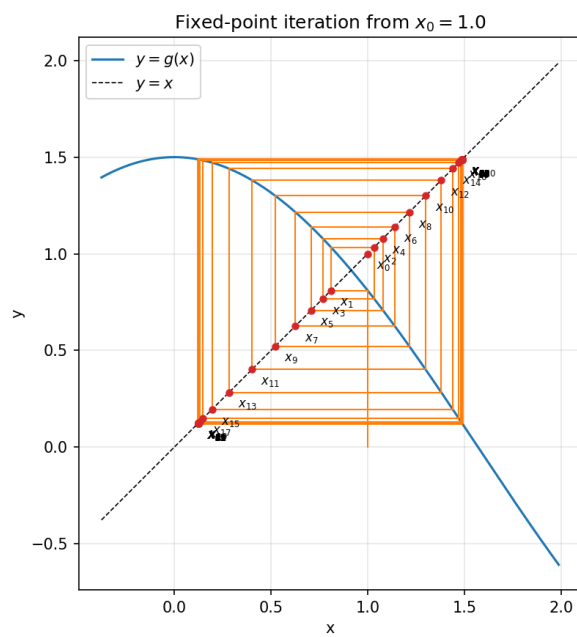
(d) x_n vs. n , $\text{tol} = 1 \times 10^{-3}$.

Remark 3.2 (Convergence not Guaranteed). [rem:fixed-point-iteration-convergence] The convergence of fixed point iteration is not guaranteed.

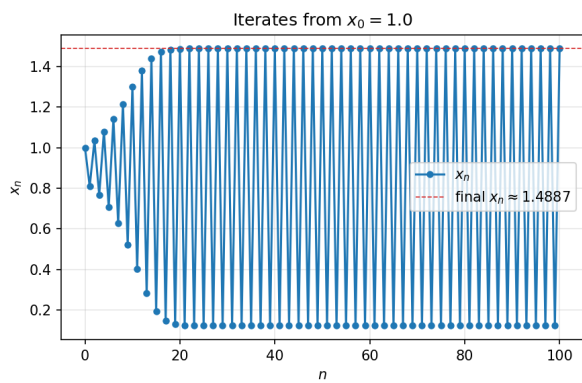
Example 3.2 (Divergent Case: $x - 1.5 \cos x = 0$). [eg:divergent-fpi] Same procedure as example 3.1, but with $g(x) = 1.5 \cos x$ and $x_0 = 1$. It diverges and oscillates between ≈ 0.123 and ≈ 1.489 ; both runs hit $N = 100$ without ever satisfying the stop condition.



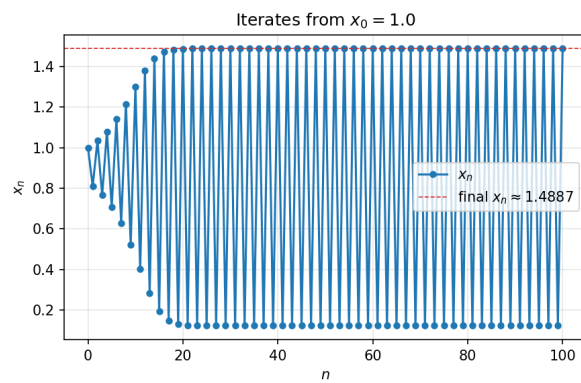
(a) Cobweb, $\text{tol} = 5 \times 10^{-2}$.



(b) Cobweb, $\text{tol} = 1 \times 10^{-3}$.



(c) x_n vs. n , $\text{tol} = 5 \times 10^{-2}$.



(d) x_n vs. n , $\text{tol} = 1 \times 10^{-3}$.